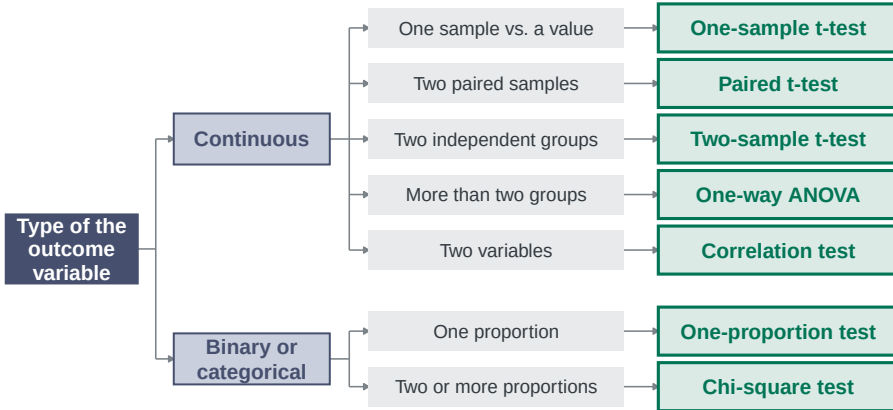


Choosing a Test

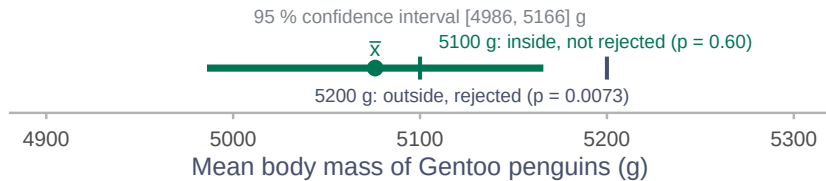
Addendum to Unit 2, 6 October 2026

A decision guide



Two questions select the test: What type is the outcome, and what is the design?

One sample against a value



- Typical use: A continuous outcome is compared with a value from outside the study, such as a norm value or a value from the literature. Test: The **one-sample t -test**.
- Gentoo body mass against 5200 g: The value lies outside the 95 % interval [4986, 5166] g and is rejected ($p = 0.0073$). In R: `t.test(x, mu = 5200)`.

Two independent groups

Two-sample t -test, Welch's version

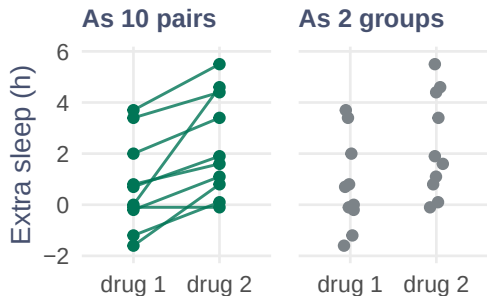
Typical use: A continuous outcome compared between two independent groups, such as two species or a treated and a control group. It compares the two means; R uses Welch's version by default.

Example (Flipper length of Adelie and Chinstrap penguins)

Chinstrap flippers are on average 5.87 mm longer (95 % confidence interval 3.86 to 7.88 mm), $p = 6.0 \cdot 10^{-8}$. In R: `t.test(x, y)`.

Assumptions: Independent units, and in each group roughly normal values or a large sample.

Two paired samples

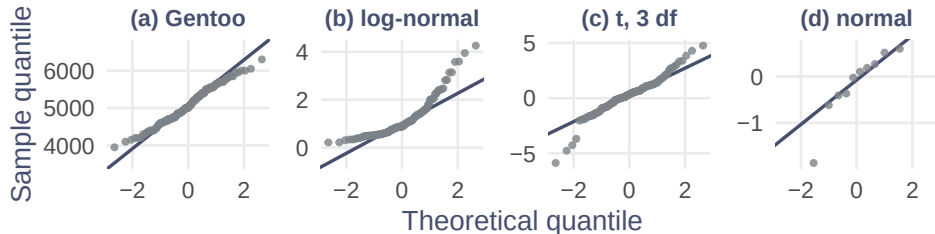


Example (sleep: Extra sleep of ten patients under two drugs)

Drug 2 – drug 1: Mean difference 1.58 h, 95 % confidence interval 0.70 to 2.46 h, $p = 0.0028$. The same values as two independent groups: $p = 0.079$.

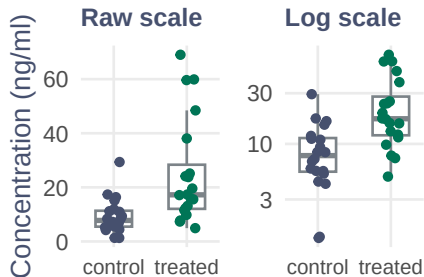
- Typical use: The same unit is measured twice, under two conditions or before and after a treatment. Test: The **paired t -test**, a one-sample t -test of the differences.
- In R: `t.test(x, y, paired = TRUE)`.

Are the values roughly normal?



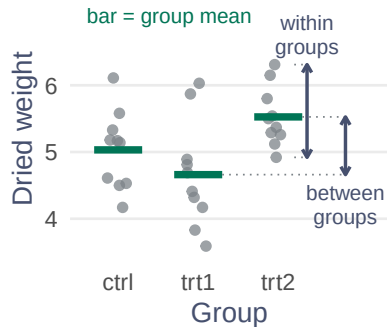
Normal quantile plots: (a) Body mass of the 123 Gentoo penguins, (b) to (d) simulated samples. For normal data the points lie close to the line. In R:
`qqnorm(x); qqline(x).`

If not: Logarithm or ranks



1. Mild deviation and n not very small:
The t -test is fine (central limit theorem).
2. Right-skewed positive values such as concentrations: Analyse the logarithm; the effect is then a ratio.
3. Very small n or extreme values: A rank-based test, such as the **Wilcoxon rank-sum test** for two groups, in R `wilcox.test(x, y)`.

More than two groups: One-way ANOVA

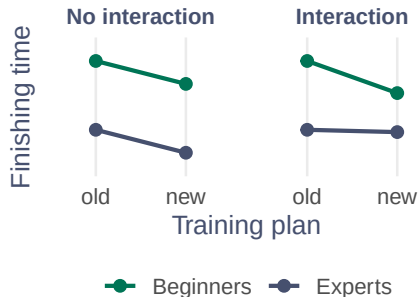


PlantGrowth: Dried weight of 30 plants, ten each in a control group and two treatment groups. The **one-way analysis of variance** compares how far the group means lie apart (between) with how much the plants scatter around their group mean (within).

- Result: $F = 4.85$, $p = 0.016$, so the three means are not all equal. Tukey's test then shows which pairs differ: Only treatment 2 against treatment 1.
- Three separate t -tests at the 5 % level would give about a 12 % chance of at least one false positive. In R: `aov(y ~ group)`, then `TukeyHSD()`.

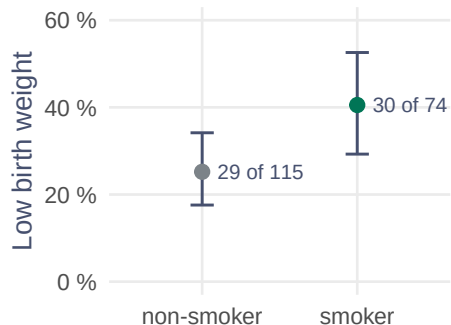
Two factors: Two-way ANOVA

Finishing times of beginners and experts under the old and the new training plan, in two constructed scenarios. Without interaction the lines are parallel, and the new plan saves the same time in both groups. With interaction the new plan helps the beginners only.



- Typical use: Two factors in one study and the question whether the effect of one depends on the other. In R: `summary(aov(time ~ plan * experience))`.
- The output has one row per factor and one for the interaction, `plan:experience`. Read the interaction row first: If it is significant, report the effect of one factor for each level of the other.

Counts: The chi-square test



Typical use: A binary or categorical outcome in two or more groups. The **chi-square test** compares the observed counts with the counts expected if the groups did not differ.

- `MASS::birthwt`: Low birth weight in 30 of 74 smokers (40.5 %) and 29 of 115 non-smokers (25.2 %);
`chisq.test(tab)`: $p = 0.040$
- If an expected count is below 5:
Fisher's exact test,
`fisher.test(tab)`

Tests and their R functions

outcome	design	if the assumptions hold	if they are doubtful
continuous	one sample vs. a value	<code>t.test(x, mu =)</code>	<code>wilcox.test(x, mu =)</code>
continuous	two independent groups	<code>t.test(x, y)</code> , Welch	<code>wilcox.test(x, y)</code> , or the log scale
continuous	paired samples	<code>t.test(x, y, paired = TRUE)</code>	<code>wilcox.test(x, y, paired = TRUE)</code>
continuous	more than two groups	<code>aov()</code> , then <code>TukeyHSD()</code>	<code>kruskal.test(y ~ g)</code>
continuous	two factors	<code>aov(y ~ A * B)</code>	transformation, such as the log scale
binary	one sample vs. a value	<code>binom.test()</code> , exact	–
binary	two groups	<code>prop.test()</code> or <code>chisq.test()</code> , same p	<code>fisher.test()</code> if an expected count is small
categorical	$r \times c$ table	<code>chisq.test()</code>	<code>fisher.test()</code>
continuous	two variables, same units	<code>cor.test(x, y)</code>	the same with <code>method = "spearman"</code>

All tests in the table assume independent units, for example one value per animal; of repeated measurements, only pairs are covered here.

Further tests

situation	test	in R
the same units under three or more conditions, values not normal	Friedman test	<code>friedman.test()</code>
the same units under three or more conditions, roughly normal	repeated-measures ANOVA, mixed model	<code>lmer()</code> from <code>lme4</code>
a binary outcome measured twice on the same units, before and after	McNemar test	<code>mcnemar.test()</code>
several 2×2 tables, for example one per hospital	Cochran-Mantel-Haenszel test	<code>mantelhaen.test()</code>
time until an event, some units without the event	log-rank test	<code>survdifff()</code> from <code>survival</code>
the whole distribution of two groups, not only the location	Kolmogorov-Smirnov test	<code>ks.test()</code>
equal variances in several groups	Levene test	<code>leveneTest()</code> from <code>car</code>
showing that two means are practically equal	equivalence test (TOST)	two one-sided <code>t.test()</code>

These tests were not part of the course. They follow the same logic: Hypotheses, a test statistic, and a p-value, reported with an estimate and its interval.

Rules of thumb for choosing a test

1. Choose the test from the type of outcome and the design.
2. The same unit measured twice: A paired test. Repeated measurements of one unit are not independent.
3. Continuous and roughly symmetric: t -test or ANOVA. Right-skewed positive values: The logarithm. Very small n or extreme values: A rank test.
4. More than two groups: One ANOVA, then Tukey's test for the pairs, not many t -tests.
5. Counts: The chi-square test; if an expected count is below 5, Fisher's exact test.
6. Many tests raise the chance of a false positive: 20 tests at the 5 % level give 64 %. Decide in advance which comparisons answer your question.
7. Report the estimate with its confidence interval and n , not only p . A non-significant result does not show that there is no difference.